MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT CONTROLLER OF MINES UNDER GEOLOGY & MINING, COMMERCE & INDUSTRIES DEPARTMENT, GOVERNMENT OF MIZORAM.

(MARCH-2026)

TECHNICAL PAPER-I

(Use of Scientific calculator is permitted)

Time Allowed : 2 hours FM : 200

PART - A (Multiple Choice Questions) (100 Marks)

All questions carry equal mark of 2 each. Attempt all questions.

*This section should be answered only on the **OMR Response Sheet** provided.*

1. Contiguous seams mean the parting between two seams is within:

(a) 10 m (b) 9 m

(c) 8 m (d) 12 m

2. Given the following:

Mining method	Technique
P: Thick seam extraction	1: Double unit face
Q: Bord and pillar extraction	2: Jet cutting
R: Longwall face development	3: Inclined slicing
S: Hydraulic mining	4: Half-moon method

The correct match is:

(a) P-3. Q-4, R-2, S-1 (b) P-3. Q-2, R-1, S-4

(c) P-1, Q-4, R-2, S-3 (d) P-3, Q-4, R-1, S-2

3. Given the following:

Excavating/loading machine	Transportation scheme
P: Bucket Wheel Excavator	1: Mine tub
Q: Continuous Miner	2: Armoured flexible chain conveyor
R: Shearer	3: Shiftable Conveyor
S: Load Haul Dumper	4: Shuttle car

The correct match is:

(a) P-3, Q-2, R-4, S-1 (b) P-3, Q-4, R-2, S-1

(c) P-3, Q-2, R-1, S-4 (d) P-1, Q-4, R-3. S-2

4. The opening cut in opencast mining is known as:

(a) berm (b) box cut

(c) trench (d) drift

5. Adit is adopted for:

(a) a deposit having less depth of cover (b) hilly terrain

(c) a deposit having more depth of cover (d) quick return on investment

6. Size of pillars is decided based on:

(a) nature of roof and floor (b) time dependent strain

(c) depth from surface (d) coal strength

7. Why is fleet angle limited to 1.5°?

(a) to reduce the side travel on the pulley

(b) to prevent excessive wear & tear of the winding rope

(c) it allows maximum persons to travel at a time in a cage

(d) it allows use of rope having lesser diameter

8. What percentage of (CO formed/O_2 absorbed) ratio indicates existence of heating?

(a) 0.5 % (b) 1%

(c) 2% (d) 3%

9. Which of the following factor is considered for calculation of cycle time of excavating machine such as shovel, drag line etc.?

(a) bucket factor (b) swell factor

(c) swing factor (d) drill factor

10. Prop density is defined as:

(a) weight carried by a single prop

(b) time elapsed to set a prop

(c) number of props required to support one sq. meter area

(d) number of persons required to erect a prop

11. What is the minimum length of roof bolt used in underground coal mine?

(a) half of the width of roadway (b) one-third of the width of roadway

(c) equal to the width of roadway (d) double the width of roadway

12. Which of the following is known as point load index?

(a) (14 + 0.175 D) Is (b) P/D^2

(c) $\sigma m = I/3$ (d) $S = C + \sigma \tan\theta$

13. The correct order of hardness of rock in their increasing order:

(a) Gypsum < Fluorite < Quartz < Topaz (b) Fluorite < Gypsum < Quartz < Topaz

(c) Fluorite < Quartz < Gypsum< Topaz (d) Topaz < Quartz < Fluorite < Gypsum

14. Weak sandstone has modulus of elasticity of 1000 MPa and Poisson's ratio is 0.25. Its shear modulus is:

(a) 4 GPa (b) 400 Mpa

(c) 500 GPa (d) 800 GPa

15. With increase in moisture content of rock, the effect on bearing capacity of the rock:

(a) increases (b) decreases

(c) no relation with the moisture content (d) depends on the rock type

16. The grain size distribution of soil is known as:

(a) Permeability (b) Structure

(c) Porosity (d) Texture

17. In an underground working area, CH_4 and CO_2 are normally expected to be found respectively:

(a) Near the floor and along the roadway (b) Near the floor and near the roof

(c) Near the roof and near the floor (d) Along the roadway and near the floor

18. In the Coward flammability diagram, the respective percentages of methane and oxygen at the nose limit are:

(a) 14.2, 0.0 (b) 14.1, 18.2

(c) 5.8, 12.1 (d) 5.0, 19.2

19. Quantity of air to be circulated to u/g workings depends on:

(a) Manpower employed in a largest shift (b) Manpower employed in a day

(c) Total production in a day (d) Both (a) and (c)

20. Geothermic gradient is:

(a) Increase in temperature with depth. (b) Increase in pressure with depth.

(c) Decrease in temperature with depth. (d) Decrease in pressure with depth.

21. The chemical Hopcalite is used in:

(a) Self-contained breathing apparatus (b) Gas mask

(c) CO- detector (d) MSA- methanometer

22. It is the junction plane of one bed of rock with another:

(a) conformity (b) dip of bed

(c) strike of bed (d) bedding plane

23. It is a compound of $CaCO_3$ and $MgCO_3$ in various proportions.

(a) Magnetite (b) Laterite

(c) Siderite (d) Dolomite

24. If sandstone is the original rock, then what is its metamorphic rock?

(a) Slate (b) Gneiss

(c) Quartzite (d) Marble

25. In paleontology, the conversion of remains of plants and animals into rock is known as:

(a) Fossils (b) Petrification

(c) Moulds (d) Carbonization

26. A natural mineral substance of iron oxide, clay and sometimes manganese is called:

(a) Shale (b) Silt

(c) Ochre (d) Tuff

27. A coal seam in India is said to be thick when the thickness is more than:

(a) 1.5 m (b) 2.5 m

(c) 3.5 m (d) 4.5 m

28. What is the minimum size of coal pillar when the depth of coal seam from surface remains between 240 m to 360 m and gallery width does not exceed 3 m?

(a) 28 m (b) 28.5 m

(c) 29 m (d) 29.5 m

29. Protodyaknov index of Indian coals varies from:

(a) 1-4 (b) 4-8

(c) 8-12 (d) 12-16

30. Over 90% of underground coal production in India comes from:

(a) Longwall method (b) Hydraulic Mining method

(c) Board & Pillar method (d) Blasting Gallery method

31. The suitable method of mining a 2 m thick strong sulphide ore deposit within strong wall rocks dipping at 18° and occurring at 60 m below the surface is:

(a) Overhand stoping (b) Top slicing

(c) Underhand longwall stoping (d) Breast stoping

32. In-situ ore of 2.0 % Cu got diluted with waste rock of value 0.4 Cu. If the resultant extracted ore has 1.6 % Cu, what is the percentage of dilution?

(a) 18% (b) 20%

(c) 25% (d) 30%

33. As per DGMS guidelines for soft ground (alluvium, gravel, clay, debris), the maximum bench height shall not exceed:

(a) 1.0 m (b) 1.5 m

(c) 2.0 m (d) 3.0 m

34. Factor of Safety (FOS) of a slope when there is a tension crack and water pressure present is:

(a) $FOS=\frac{c.\ A+(W.\ Cos\beta\ -V)tan\varphi}{W.\ Sin\beta+U}$ (b) $FOS=\frac{c.\ A+(W.\ Cos\beta\ +V)tan\varphi}{W.\ Sin\beta\ -\ U}$

(c) $FOS=\frac{c.\ A+(W.\ Sin\beta\ -V)tan\varphi}{W.\ Cos\beta\ +\ U}$ (d) $FOS=\frac{c.\ A+(W.\ Sin\beta\ +V)tan\varphi}{W.\ Cos\beta\ -\ U}$

35. Match the following features with the corresponding symbols.

	Feature in mine plan		Symbol
P	Shaft	1	
Q	Staple shaft	2	
R	Abandoned shaft	3	
S	Abandoned staple shaft	4	

(a) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$ (b) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

(c) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$ (d) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

36. _____ material has different property in different axes.

(a) Isotropy (b) Anisotropy

(c) Angiostropy (d) Monotropy

37. The void ratio of an unconsolidated soil heap of volume 1000 m^3 is 1.0. If the soil heap is consolidated to a volume of 800 m^3, the corresponding void ratio is:

(a) 0.6 (b) 0.5

(c) 1 (d) 1.5

38. The angle between the major principal stress (σ_1) and vertical stress (σ_z) is given by:

(a) $\theta_z = \frac{1}{2} Cos^{-1}\left(\frac{2\sigma z - \sigma 1 - \sigma 3}{\sigma 1 - \sigma 3}\right)$ (b) $\theta_z = \frac{1}{2} Cos^{-1}\left(\frac{2\sigma z + \sigma 1 + \sigma 3}{\sigma 1 + \sigma 3}\right)$

(c) $\theta_z = \frac{1}{2} Tan^{-1}\left(\frac{2\sigma z - \sigma 1 - \sigma 3}{\sigma 1 - \sigma 3}\right)$ (d) $\theta_z = \frac{1}{2} Tan^{-1}\left(\frac{2\sigma z + \sigma 1 + \sigma 3}{\sigma 1 + \sigma 3}\right)$

39. In the generalized Hoek–Brown equation, the uni-axial compressive strength of a rock mass is obtained by setting:

(a) $\sigma_1 = 0$ (b) $\sigma_2 = 0$

(c) $\sigma_3 = 0$ (d) $m_i = 0$

40. For a cylindrical specimen of cross section area A, length L, Young's modulus E, and axial load P, the strain energy stored is:

(a) $\frac{P^2 L}{2\,A.E}$ (b) $\frac{P.L^2}{2\,A.E}$

(c) $\frac{P^2.L^2}{A.E}$ (d) $\frac{P^2 L}{A.E}$

41. The oxygen concentration in underground mines should not be less than:

(a) 18% (b) 19%

(c) 20% (d) 21%

42. A coal mine with methane emission rate $\leq$ 1 m^3/tonne is:

(a) Degree I (b) Degree II

(c) Degree III (d) Nongassy

43. Quantity of air to be circulated to U/G working where 150 workers employed in the largest shift and daily production is 500 tonnes will be:

(a) 950 m^3/min (b) 1050 m^3/min

(c) 1150 m^3/min (d) 1250 m^3/min

44. In an underground coal mine, a Graham's Ratio of 3% is an indication of:

(a) Existence of spontaneous heating (b) Active fire

(c) Heating in advance stage (d) Normal coal mine

45. The early indication of spontaneous heating of coal can be given by:

(a) Sweating of strata (b) Formation of CO^2

(c) Rise of temperature (d) Smoke from the coal face

46. The hardness of a mineral depends mainly on:

(a) Cleavage (b) Luster

(c) Colour (d) Bond strength between atoms

47. Dip is measured:

(a) Vertically (b) Horizontally

(c) Perpendicular to Strike (d) Parallel to Strike

48. A fold with both limbs dipping toward the axis is:

(a) Anticline (b) Syncline

(c) Monocline (d) Recumbent Fold

49. The era known as the "Age of Reptiles" is:

(a) Mesozoic (b) Paleozoic

(c) Cenozoic (d) Precambrian

50. Limestone deposits in India are extensively used for:

(a) Fertilizers (b) Steel Industries

(c) Cement Industries (d) Medicinal purpose

PART- B *(Conventional Type) (100 Marks)*

*This section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it. Attempt all questions.

1. Define the following terms: **(4×1=4)**

(a) Gestation Period
(b) Block Caving
(c) Stripping Ratio
(d) Break – Even Stripping Ratio

2. In an Under Ground Coal Mine, 1800 workers are employed. The average production from the mine is 900 tonne per day. Assuming 3 working shifts per day, calculate the Output per Man Shift (OMS). **(5)**

3. Broken rock of 3684 m^3 is heaped into a square pyramid of height 12 m. Calculate the base area in metre square. **(5)**

4. Four benches are being worked by the opencast mining system. Height, width and face angle for each bench are 15m, 50m and 70º respectively. Calculate the overall slope angle of the benches. **(5)**

5. The ratio of uniaxial compressive strength to uniaxial tensile strength of a sandstone specimen is 8:1. Calculate the theoretical value of angle of internal friction of the specimen in degree. **(5)**

6. A series of triaxial tests of sandstone samples reveal the cohesion and the angle of internal friction as 21.65 MPa and 30º respectively. Based on the assumption that the sandstone samples and by following the Mohr-Coulomb's failure criteria, calculate the tensile strength in Mpa. **(6)**

7. In a mine ventilation system, the resistances of two splits A and B are 0.5 Ns^2m^{-8} and 2 Ns^2m^{-8} respectively. Combined resistance of two shafts and trunk airways is 0.7 Ns^2m^{-8}. A quantity of 20 m^3/s of air passes through split A. Calculate the total air quantity passing the mine in m^3/s. **(5)**

8. State the three fundamental Fan Laws used in mine ventilation. If the rotational speed of a mine fan is doubled, what will be the resulting effect on the quantity of airflow, ventilating pressure, and fan power? **(5)**

9. Define dip and strike. Illustrate with suitable diagram. **(5)**

10. Define Fold? Distinguish between Anticline and Syncline by illustrating with the help of suitable diagrams. **(5)**

11. Define *Ultimate Analysis* and *Proximate Analysis* of coal. List the constituents determined in each method. **(5)**

12. In an underground coal mine using the bord and pillar method, a development panel provides the following production data for a single shift: **(5)**

Number of blasting rounds fired per shift: 8

Dimensions of the face exposed for each blast: 4.2 m (width) × 2.5 m (height)

Average pull obtained per blast: 1.2 m

Specific gravity of coal: 1.4

Total manpower employed in the panel per shift: 100

(a) Determine the Output per Manshift (OMS) for the panel in tonnes.

(b) Comment briefly on whether the OMS value indicates high, moderate, or low productivity for a development district in an underground coal mine.

13. A rectangular coal pillar is formed with center-to-center dimensions of 30 m × 30 m in an underground mine. The width of each gallery is 5.0 m, and the depth of the working is 200 m. If the unit weight of the overlying strata is 0.025 MPa/m, calculate the load (stress) acting on the pillar using the Tributary Area Method. **(5)**

14. What is Rock Quality Designation (RQD)? Explain how it is determined and give the standard classification of rock mass based on RQD values. **(5)**

15. Rock mass parameters for a shale rock, m_β, s, and a of the Hoek and Brown failure criterion as given in the equation are 2.4, 0.01 and 0.52 respectively. The uni-axial compressive strength (σ_{ci}) of intact shale is 50 MPa.

$$\sigma_1 = \sigma_3 + \sigma_{ci}(m_\beta \cdot \sigma_3 / \sigma_{ci} + s)^a$$

Find the uniaxial compressive strength of rock mass. **(5)**

16. Define Modulus of Elasticity of rock. Derive the expression $E=\frac{\sigma}{\varepsilon}$ and explain the difference between Tangent modulus and Secant modulus. **(5)**

17. Sample of air collected in the intake and return gate roads of an advancing longwall face show 0.25% and 0.75% CH_4, respectively. The panel has a production of 400 tonnes in an 8-hour shift, when an air quality of 18 m^3/s passes through it, what will be the emission in m^3/t of coal produced. **(5)**

18. Calculate the N.V.P in an underground mine using the following data: **(5)**

Depth of mine = 300 m

Pit bottom barometer reading = 101.27 kPa

Pit-top barometer reading = 98.10 kPa

Average temperature in downcast shaft = 307 K

Average temperature in upcast shaft = 304 K

19. What are joints in rocks? Classify different types of joints and discuss their effect on blasting and excavation workability. **(5)**

20. Classify mineral deposits based on their mode of origin. Describe any three types with suitable examples. **(5)**

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